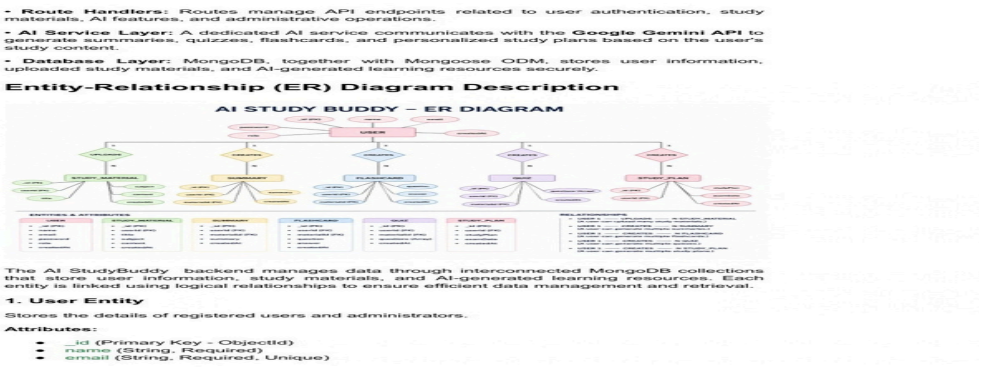
LEARNMATE – AI STUDYBUDDY

7. Project Documentation Phase

Documenting the architecture, entities, features, roles, APIs and execution process.

Date	21-09-26
Team ID	SWTID-2026-5714
Project Name	LearnMate - AI StudyBuddy
Maximum marks	4 Marks



- **Route Handlers:** Routes manage API endpoints related to user authentication, study materials, AI features, and administrative operations.
- **AI Service Layer:** A dedicated AI service communicates with the **Google Gemini API** to generate summaries, quizzes, flashcards, and personalized study plans based on the user's study content.
- **Database Layer:** MongoDB, together with Mongoose ODM, stores user information, uploaded study materials, and AI-generated learning resources securely.

Entity-Relationship (ER) Diagram Description

The AI StudyBuddy backend manages data through interconnected MongoDB collections that store user information, study materials, and AI-generated learning resources. Each entity is linked using logical relationships to ensure efficient data management and retrieval.

1. User Entity

Stores the details of registered users and administrators.

Attributes:

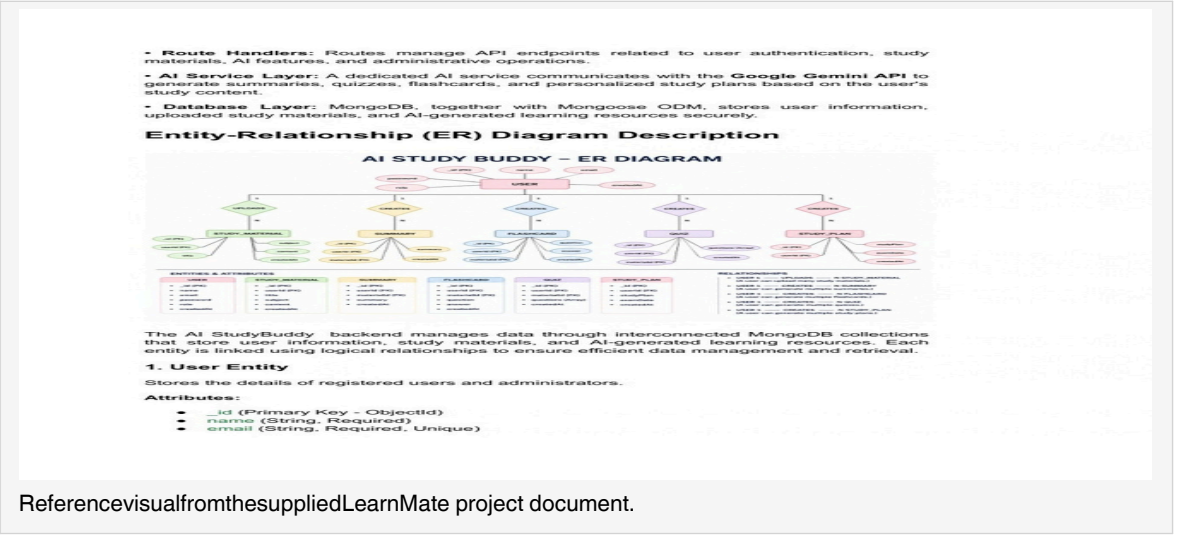
- _id (Primary Key - ObjectId)
- name (String, Required)
- email (String, Required, Unique)

AI-powered learning assistance and smart study support platform

System Documentation

Key Points

- Document the modular RESTful architecture and separation of authentication, business logic, AI services and database operations.
- Document the User, Study Material, Summary, Flashcard, Quiz and Study Plan entities.
- Record the relationships between users, study materials and generated learning resources.


- **Route Handlers:** Routes manage API endpoints related to user authentication, study materials, AI features, and administrative operations.
- **AI Service Layer:** A dedicated AI service communicates with the **Google Gemini API** to generate summaries, quizzes, flashcards, and personalized study plans based on the user's study content.
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ReferencevisualfromthesuppliedLearnMate project document.

Feature Documentation

Key Points

- Secure User Authentication using JWT and bcrypt.
- Study Material Management for uploaded learning content.
- AI Summary, Flashcard, Quiz and Study Plan generation using Gemini.
- Role-Based Access Control for Student and Administrator roles.
- MongoDB data storage for users, materials and generated resources.

User & Admin Roles

Key Points

- Students register, log in, upload materials, generate AI resources and manage previously generated resources.
- Administrators manage users, monitor application usage, maintain system integrity and oversee AI services.

Documentation Outcome

Key Points

- The documentation provides a structured record of how the system is designed, developed, executed and accessed through REST APIs.